Ernest Uwaoma

Software Engineer (Full Stack)

Lagos, Nigeria (open to remote worldwide) | ernestuwaoma3@gmail.com | +234 803 300 4474
ernestuwaomaportfolio.vercel.app | github.com/Nexkeys | linkedin.com/in/ernest-uwaoma-183338344 | sellapage.com.ng

PROFILE

Full stack software engineer with 6 years of shipping production software for paying clients and for my own products across commerce, fintech, real-time gaming and developer tools. I own systems end to end: I write the architecture decisions, design the data model and APIs, set up cloud infrastructure and CI, and run what I ship in production. Most recently I planned and carried out Sellapage's move from a Firebase client app to a typed Nx monorepo with a Next.js web app, one Node.js API on Google Cloud Run and PostgreSQL, all managed in Terraform. Strongest in TypeScript, React, Next.js, Node.js and PostgreSQL, with deep hands-on work in payments, real-time systems and practical LLM features.

TECHNICAL SKILLS

Languages	TypeScript, JavaScript, Python, SQL, Java, HTML, CSS, MQL5
Frontend	React 19, Next.js 16 (App Router), React Router 8, Astro, React Native and Expo, Tailwind CSS, Three.js and React Three Fiber, GSAP, Motion, PWAs
Backend and data	Node.js, Express, REST API design, WebSockets, PostgreSQL (Cloud SQL), Firebase (Auth, Firestore, Realtime Database), Supabase, MongoDB, MySQL, SQLite (including sql.js in the browser), Zod
Cloud and DevOps	Google Cloud (Cloud Run, Cloud SQL, Secret Manager, Artifact Registry, Cloud Build, Cloud Monitoring), Terraform, Docker, GitHub Actions, Workload Identity Federation, Nx and pnpm monorepos, Vercel, Netlify, Firebase Hosting, Azure
Testing and quality	Vitest, Jest, Mocha, Playwright, Firebase Emulator Suite and security rules tests, ESLint, TypeScript strict mode, visual regression checks
AI and dev tools	OpenAI, Anthropic, Groq, Gemini and Ollama APIs, LLM routing with offline fallback, validated natural language to SQL, Model Context Protocol (MCP) servers, VS Code Extension API, Electron, ts-morph
Payments and logistics	Paystack (webhooks, idempotency, reconciliation), Flutterwave, Sendbox, Topship, Terminal Africa, DHL
Product and writing	UX writing and microcopy, Figma, Postman, Architecture Decision Records, runbooks

EXPERIENCE

Founder and Software Engineer, Sellapage

Apr 2026 to Present | Remote

All in one commerce platform for Nigerian merchants, on the web and on Google Play | sellapage.com.ng

- Built the product solo from zero to a live first version in 26 days: storefront builder, Paystack checkout, Sendbox and Topship delivery, customer CRM, bookings, referrals, analytics, job listings, a blog and an AI assistant (Sella). It has grown to 170+ active merchants, 200+ users and over NGN 500,000 in payment volume with no paid acquisition.
- Planned and led a 15 phase technical roadmap, documented in 10 Architecture Decision Records: moved the product from a Firebase client app to an Nx and pnpm monorepo with a Next.js 16 web app, one Node.js API service on Cloud Run and PostgreSQL 18 on Cloud SQL, then retired Firestore completely (October 2026) once every read and write went through the API.
- Made the trade-offs that kept the migration within one engineer's capacity without freezing feature work: moved data first and reshaped it later, and chose one API service over one service per domain, each recorded with the alternatives and reasons.
- Wrote the Google Cloud foundation as Terraform modules: three projects, private Cloud SQL with IAM sign in and point in time recovery, Secret Manager, Artifact Registry, alerts, uptime checks and a spending budget; replaced long lived deploy keys with GitHub Workload Identity Federation.
- Built the financial core: payment intents, reconciliation and a ledger, with revenue counted only from Paystack-confirmed payments and idempotent webhook handlers so retried events never credit twice. Fixed and tested 16 priority security and payment findings before the migration began.
- Set up CI on every pull request (build, blocking lint, 130+ test files, emulator rules tests, visual comparisons, secret and dependency scans) and shipped the Android app (Expo, React Native) on the same API.

Founder and Lead Engineer, NexKeys Agency

Jul 2020 to Present | Remote

Software studio delivering web, mobile and AI products for clients in Nigeria and abroad | nexkeysagency.com.ng

- Shipped 12+ production products for 13+ clients across fintech, gaming, commerce and civic tech, with repeat engagements from several clients; I own each project from the first brief to post-launch support.
- Turn loose briefs into written technical specs, choose the architecture and explain the trade-offs to non-technical clients; review code from contract engineers and set the branching, pull request review and GitHub Actions deployment conventions that cut production incidents across engagements.
- Built JWT-secured REST APIs with tiered rate limiting, and Paystack and mobile money integrations with fraud checks, across several client platforms.

- Delivered production UX microcopy for four FSDH Merchant Bank digital properties (Asset Management, Holdco, Merchant Bank, FSDH Capital) in a scoped contract sprint, shipped in a single deployment cycle.

Selected client work

- **Valor Perks** (2024 to 2025; React, Node.js, PostgreSQL): token trading platform with real-time token valuation, automated payouts and encrypted payment flows with fraud prevention that reduced failed transactions.
- **GamePlug HQ** (2025; React, Node.js, Firebase, Paystack): peer to peer gaming account marketplace with a state machine escrow (pending, locked, released, disputed) and idempotent webhooks that remove duplicate fund releases.
- **Kestrel FX Scanner and Trade Bot** (Python, MetaTrader 5 API, Docker): delivered for a paying client inside a 12 day brief; async scanner over configurable pair pools, plug-in strategies, daily loss limits, automatic stop loss and take profit, Telegram alerts.
- **Chamelioni** (React, Firebase, Tailwind CSS): furniture artisan marketplace with a built-in sketch tool, a 24 hour quote engine and artisan workflow management.
- **Jenny Wine Store** (2025; React, Tailwind CSS): responsive storefront tuned for product discovery and mobile speed; the owner reports orders tripled after launch.

SELECTED PROJECTS

NexPlay *React 19, TypeScript, React Router 8, Firebase Realtime Database, Vitest* github.com/Nexkeys/nexplay

- Real-time multiplayer platform with eight games including Ayo and Whot!, sustaining 1,000+ concurrent users at peak. Host-authoritative design: moves go to a path only the host can read, the host validates them against a pure state machine per game, then publishes a spectator view and a private view per player so no one can see another player's cards.
- Automatic host failover after 5 seconds of disconnect, commit and reveal (SHA-256) for simultaneous picks, collision-free room codes reserved with transactions, and ghost player cleanup cut from about 30 seconds to under 2.

ZapCash *TypeScript (strict), React Router 8, Firebase, sql.js (WASM), Groq, Playwright* Private repo

- Money assistant where users ask questions about their spending in plain English or Pidgin. Bank statements (CSV, PDF) are parsed on the device; the LLM receives only the question and the table schema, never transaction data.
- Generated SQL is never trusted: it is parsed, limited to one SELECT over allowed tables, given a LIMIT and a time limit, then run in SQLite inside the browser. A serverless function verifies Firebase ID tokens and enforces a daily quota; 53 test files cover parsers, the SQL checker, security rules and end to end flows on desktop and phone.

Sandbox Atlas *TypeScript, VS Code Extension API, ts-morph, SQLite, esbuild, GitHub Actions* github.com/Nexkeys/sandbox-atlas

- Open source (MIT) extension for VS Code, Cursor and Windsurf that maps a codebase locally and answers "where does this live" and "what breaks if I change this". Builds a real import graph (barrels, path aliases, ESM, require and dynamic import), so impact analysis comes from the graph rather than a model's guess.
- Pluggable providers (Ollama and LM Studio offline, or Gemini, OpenAI and OpenRouter), keys in encrypted Secret Storage, a daily cost cap for paid providers, and CI plus release workflows.

Dahl Studio *Electron, React, TypeScript, Monaco, Zustand, Vitest* github.com/Nexkeys/dahl-studio

- Local-first AI coding studio for Windows: agent turn loop with sub-agents, file leases, approvals and revert; MCP client over STDIO and HTTP; OpenAI-compatible streaming client for several providers. Keys live in a Windows DPAPI vault and are stripped from terminal environments; 43 unit tests and an NSIS installer.

Sekudo *Python, Model Context Protocol, httpx* github.com/Nexkeys/sekudo

- Security testing (DAST) server for AI coding tools with 34 checks mapped to the OWASP Top 10 (2021). A scope gate refuses any target that is not on an allow list, and every finding is proven with a non-destructive proof of concept plus a suggested fix.

CIRF *React, Express, Firebase, Zod, PDFKit, Cloudinary* github.com/Nexkeys/cirf

- Civic tech platform tracking community-funded repairs from levy to receipt, with payment checks, reconciliation and a public transparency report.

Axion AI *Python, Groq, Ollama, Vosk, openWakeWord, SQLite, ADB* github.com/Nexkeys/axion-ai

- Privacy-first voice assistant with offline wake word and speech to text, Groq with local Ollama fallback, and PC and Android control.

EDUCATION AND ADDITIONAL

Mastermind Series, Software Development, NIIT, Festac, Lagos 2025 to Present
Software Development Bootcamp, Quintex Academy, Abuja (won the coding competition) 2022 to 2024

Availability: immediately, remote worldwide, full-time or contract. **Time zones:** WAT, overlapping CET, partial US coverage.
Languages: English (C1), Igbo (native), Spanish (A1).